

$$A^{-1} = \begin{pmatrix} \frac{1}{7} & \frac{1}{7} & \frac{1}{7} \\ \frac{2}{7} & \frac{3}{14} & -\frac{2}{7} \\ -\frac{8}{7} & \frac{9}{14} & \frac{1}{7} \end{pmatrix}$$

$$b = \begin{pmatrix} 8 \\ 3 \\ 0 \end{pmatrix}$$

$$x = A^{-1}b$$

$$x = \begin{pmatrix} \frac{1}{7} & \frac{1}{7} & \frac{1}{7} \\ \frac{2}{7} & \frac{3}{14} & -\frac{2}{7} \\ -\frac{8}{7} & \frac{9}{14} & \frac{1}{7} \end{pmatrix} \begin{pmatrix} 8 \\ 3 \\ 0 \end{pmatrix}$$

$$x_1 = -\frac{5}{7}, \quad x_2 = \frac{41}{44}, \quad x_3 = -\frac{101}{14}$$

1G-2

$$A = \begin{pmatrix} 4 & 3 \\ 3 & 2 \end{pmatrix}, \quad b = \begin{pmatrix} -1 \\ 1 \end{pmatrix}$$

$$x = A^{-1}b$$

$$x = \begin{pmatrix} 5 \\ -7 \end{pmatrix}, \quad b = \begin{pmatrix} 5 \\ -6 \end{pmatrix}$$

$$x_1 = -5, \quad x_2 = -6$$

1G-3

$$M = \begin{pmatrix} 3 & -1 & 1 \\ 1 & 3 & 2 \\ -2 & -1 & 1 \end{pmatrix}$$

$$M^T = \begin{pmatrix} 3 & 1 & -2 \\ -1 & 3 & -1 \\ 1 & 2 & 1 \end{pmatrix}$$

$$A^{-1} = \frac{1}{5} \begin{pmatrix} 3 & 1 & -2 \\ -1 & 3 & -1 \\ 1 & 2 & 1 \end{pmatrix}$$

$$X = A^{-1}b$$

$$X = \frac{1}{5} \begin{pmatrix} 3 & 1 & -2 \\ -1 & 3 & -1 \\ 1 & 2 & 1 \end{pmatrix} \begin{pmatrix} 2 \\ 0 \\ 8 \end{pmatrix}$$

$$= \frac{1}{5} \begin{pmatrix} 10 \\ -5 \\ 5 \end{pmatrix} = \begin{pmatrix} 2 \\ -1 \\ 1 \end{pmatrix}$$

IG-4

The system is $Ax = y$, the solution of the equation is $x = A^{-1}y$.

$$\begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \frac{1}{5} \begin{pmatrix} 3 & 1 & -2 \\ -1 & 3 & -1 \\ 1 & 2 & 1 \end{pmatrix} \begin{pmatrix} y_1 \\ y_2 \\ y_3 \end{pmatrix}$$

Written out, this is the system of equations

$$x_1 = \frac{3}{5}y_1 + \frac{1}{5}y_2 - \frac{2}{5}y_3$$

$$x_2 = -\frac{1}{5}y_1 + \frac{3}{5}y_2 - \frac{1}{5}y_3$$

$$x_3 = \frac{1}{5}y_1 + \frac{2}{5}y_2 + \frac{1}{5}y_3$$

IG-5

Using the associative law, definition of the inverse, and identity law:

$$(B^{-1}A^{-1})(AB) = B^{-1}(A^{-1}A)B = B^{-1}IB = I$$

Similarly,

$$(AB)(B^{-1}A^{-1}) = I$$

Therefore, $B^{-1}A^{-1}$ is the inverse of AB .

MH-1

$$\begin{vmatrix} 3 & -1 & 1 \\ -1 & 2 & 1 \\ 4 & -1 & 1 \end{vmatrix} = 9 - 2 - 1 = 6$$

b)

$$|A| = \begin{vmatrix} 1 & -1 & 1 \\ 1 & 0 & -1 \\ -1 & 1 & 1 \end{vmatrix} = 2$$

$$\begin{aligned} C &= \frac{1}{|A|} \begin{vmatrix} 0 & -1 & 1 \\ 1 & 0 & -1 \\ 2 & 1 & 1 \end{vmatrix} \\ &= \frac{1}{2} - 4 = 2 \end{aligned}$$

H-2

Using Cramer's rule, the determinants in the numerators for x , y , and z all have a column of zero, therefore their value is zero.

H-3

a) The condition for it to have a non-zero solution is

$$\begin{vmatrix} 1 & -1 & 1 \\ 2 & 1 & 1 \\ -1 & c & 2 \end{vmatrix} = 0$$

Expanding,

$$2 + 2c - 11 - (-1 + c - 4) = 0$$

$$c = -8$$

b)

$$(2 - c)x + y = 0$$

$$(-1 - c)y = 0$$

A non-trivial solution exists if

$$\begin{vmatrix} 2 - c & 1 \\ 0 & -1 - c \end{vmatrix} = 0$$

$$(2 - c)(-1 - c) = 0$$

$$c = 2 \quad \text{or} \quad c = -1$$

c)

For $c = -8$, the equation gives a vector (x_1, x_2, x_3) orthogonal to the vectors

$$(1, -1, 2), \quad (2, 1, 1), \quad (-1, 8, 2)$$